Phylogenetic Analysis of the Gelsolin Protein Family Across Metazoans

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1. Objective

The objective of this study was to perform a reproducible phylogenetic analysis of the gelsolin protein family across diverse metazoan taxa. The workflow included:

- Sequence retrieval
- Dataset curation
- Multiple sequence alignment
- Alignment trimming
- Phylogenetic tree construction
- Interpretation of biological meaning

This analysis aimed to understand evolutionary conservation and diversification of gelsolin-family proteins across animal lineages.

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2. Biological Background

Gelsolin is an actin-binding protein involved in cytoskeletal remodeling and regulation of actin filament assembly/disassembly. Canonical gelsolin proteins are highly conserved in vertebrates, while more divergent gelsolin-like homologs occur in basal metazoans and invertebrates.

Because canonical gelsolin orthologs are limited outside metazoans, this analysis focused primarily on:

- Vertebrate gelsolin orthologs
- Invertebrate gelsolin-like homologs
- Basal metazoan gelsolin-domain proteins

Proteins such as villin, CapG, and Flightless-1 were excluded because they represent paralogous members of the broader gelsolin superfamily and could confound ortholog-focused phylogenetic inference.

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3. Sequence Retrieval

Database Used

Sequences were retrieved primarily from UniProt.

Website: https://www.uniprot.org/

Criteria for inclusion

Sequences were included if they:

- belonged to the gelsolin/gelsolin-like family
- contained conserved gelsolin domains
- were sufficiently full-length
- aligned reasonably with other homologs
- represented evolutionarily informative taxa

Criteria for exclusion

Sequences were excluded if they:

- were partial fragments
- were excessively truncated
- represented unrelated multidomain proteins
- corresponded to villin, CapG, or Flightless-1
- showed extremely poor alignment quality

Taxonomical diversity was maintained in dataset to capture any taxa specific relationship in sequences and also compare and visualize the evolutionary changes in sequence.

The final dataset included representatives from:

- Basal metazoans
- Invertebrates
- Chordates
- Vertebrates

Representative taxa included:

- Homo sapiens
- Mus musculus
- Rattus norvegicus
- Gallus gallus
- Xenopus laevis

- Danio rerio
- Drosophila melanogaster
- Anopheles gambiae
- Caenorhabditis elegans
- Brugia malayi
- Nematostella vectensis
- Amphimedon queenslandica
- Trichoplax adhaerens
- Octopus bimaculoides
- Strongylocentrotus purpuratus

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4. FASTA Dataset Preparation

All curated sequences saved in “raw_seq” file were combined into a single FASTA file using Python.

Combined dataset file:

combined_gelsolin.fasta

Script used is uploaded in file “codes used” and named “combine_fasta.py”

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5. Multiple Sequence Alignment (MSA)

Software Used

MAFFT

Website: https://mafft.cbrc.jp/alignment/software/

Command: mafft –auto combined_gelsolin.fasta > aligned.fasta

Parameters

–auto

MAFFT automatically selected the most suitable alignment algorithm based on dataset size and complexity.

Output

msa.fasta

6. Alignment Trimming

Software Used

ClipKIT

Website: https://github.com/JLSteenwyk/ClipKIT

Purpose

Alignment trimming was performed to:

- remove poorly aligned regions
- eliminate highly gapped positions
- improve phylogenetic signal
- reduce noise during tree construction

Command Used

python -m clipkit aligned.fasta -o trimmed.fasta

Output

trimmed.fasta

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7. Phylogenetic Tree Construction

Software Used

IQ-TREE v2

Website: https://iqtree.github.io/

Command Used

iqtree2.exe -s trimmed.fasta -m TEST -bb 1000 -alrt 1000

Parameters

-s trimmed.fasta

Input trimmed alignment.

-m TEST

Automatic evolutionary model selection.

-bb 1000

Ultrafast bootstrap approximation with 1000 replicates.

-alrt 1000

SH-aLRT branch support with 1000 replicates.

Output files are saved in folder “iq tree files”

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8. Tree Visualization

The names of the organisms was changed to more readable format before tree visualization .

Example :

tr|A0A0L8GZ50|A0A0L8GZ50_Octopus_bimaculoides ---→ Octopus_bimaculoides

Script used is uploaded as “species_names_only.py”

Software Used

iTOL (Interactive Tree Of Life)

Website: https://itol.embl.de/

Visualization Procedure

The following file was uploaded:

trimmed.fasta.treefile

Species labels were cleaned to replace accession identifiers with readable organism names.

The tree was visualized in rectangular format and exported as PDF.

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9. IqTree Result analysis

Alignment Statistics

- 25 sequences
- 1105 alignment positions
- 754 parsimony-informative sites

Indicated strong phylogenetic signal.

Best-fit Evolutionary Model

IQ-TREE selected:

Q.PFAM+I+G4

according to Bayesian Information Criterion (BIC).

Bootstrap Convergence

The bootstrap correlation coefficient reached:

1.000

indicating stable and converged bootstrap estimates.

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10. Biological Interpretation

Evolutionary Trends

The phylogenetic tree revealed three major regions based on evolution:

1. Basal divergent metazoan gelsolin-like proteins
2. Intermediate invertebrate homologs
3. Highly conserved vertebrate gelsolin orthologs- tightly packed

Basal Metazoan Divergence

Basal taxa including Trichoplax adhaerens, Strongylocentrotus purpuratus, Octopus bimaculoides formed highly divergent group with longer branch lengths.

This suggests early diversification of gelsolin-family proteins during metazoan evolution.

Invertebrate Clustering

Insect and nematode sequences formed biologically consistent clusters.

Examples: Drosophila melanogaster clustered with Anopheles gambiae, Caenorhabditis elegans clustered with Brugia malayi

This supports retention of evolutionary signal within the data.

Vertebrate Conservation

Vertebrate sequences formed a highly conserved clade.

Mammals clustered tightly:

- Homo sapiens
- Mus musculus
- Rattus norvegicus
- Bos taurus
- Canis lupus

Birds and fish also formed expected lineage-specific groupings.

This suggests strong conservation of gelsolin function across vertebrates.

Functional Interpretation

The strong conservation of vertebrate gelsolin proteins likely reflects essential actin-binding function, conserved cytoskeletal regulation across taxa.

Basal taxa showed increased divergence, more gap frequencies in msa and composition heterogeneity, suggesting: functional specialization, evolution of gelsolin domain evolution, adaptation of cytoskeletal systems during early metazoan diversification and coarse of evolution.

11. Reproducibility

All datasets, alignments, scripts, and tree files were retained for reproducibility.

Repository structure included:

- raw FASTA sequences
- curated FASTA datasets
- alignment files
- trimmed alignments
- IQ-TREE outputs
- scripts used for header cleaning and file preparation
- visualization outputs

Conclusion

This study successfully reconstructed the evolutionary relationships of gelsolin-family proteins across diverse metazoan taxa using a reproducible maximum-likelihood phylogenetic workflow.

The analysis demonstrated deep divergence of basal metazoan gelsolin-like proteins

lineage-consistent clustering of invertebrate homologs, strong conservation of vertebrate gelsolin orthologs

Overall, the results support progressive evolutionary stabilization and specialization of gelsolin-mediated cytoskeletal regulation during metazoan evolution.